



Data Converters

Chapter 08 – Delta-Sigma A/D Converters

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Acknowledgement

Concepts & slides are partially based on the lecture

VLSI Data Conversion Circuits

held by

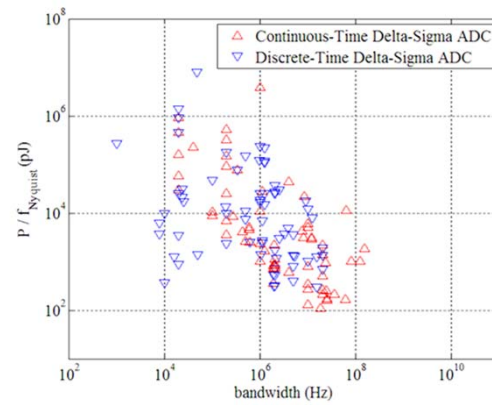
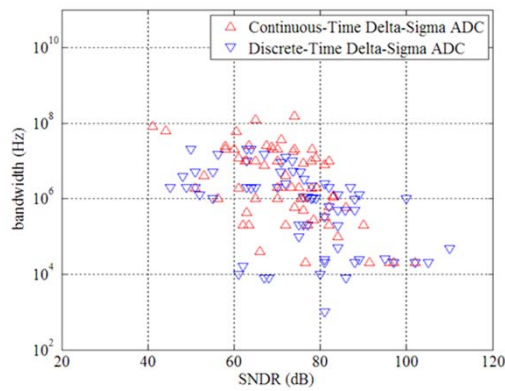
Prof. B. Murmann, PhD, University of Hawai'i
during his sabbatical at the University of Freiburg in 2009



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State Of The Art – Overview



based on the data provided at
B. Murmann, "ADC Performance Survey 1997-2013," [Online]
available: <http://www.stanford.edu/~murmman/adcsurvey.html>

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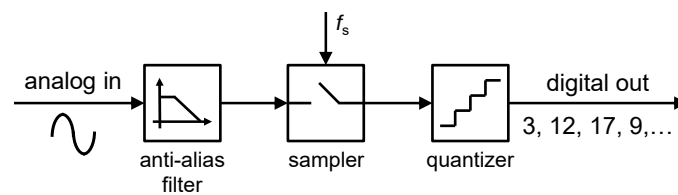
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Recollect...

A/D conversion



sampling theorem

$$f_{s,\min} = 2f_{\text{sig,max}} = 2f_b$$

in practice slightly oversampled, i.e.,

$$f_s > 2f_{\text{sig,max}} = 2f_b$$

→ relaxes specifications of anti-alias filter

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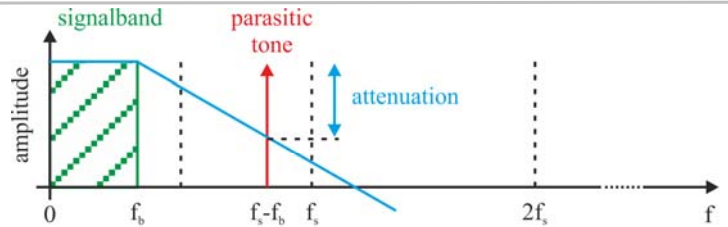
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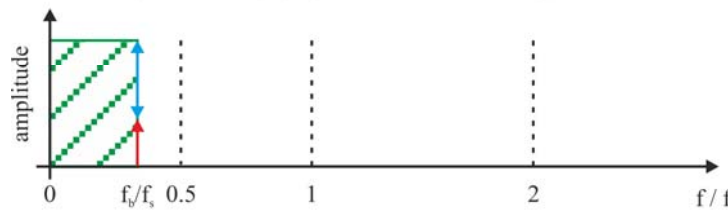


Anti-Aliasing Filter

continuous time



discrete time



Sampling rate faster than Nyquist required for adequate attenuation!

→ "Oversampling"

→ trade off sampling rate ↔ filter order

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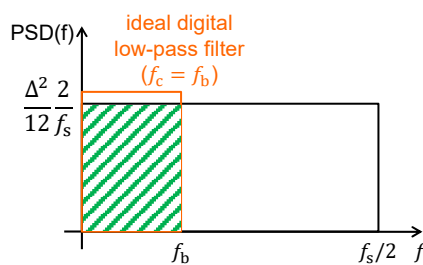
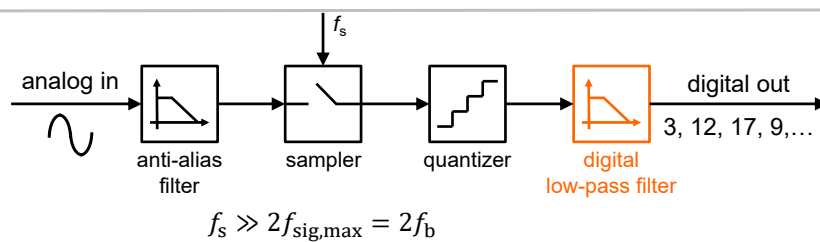
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Oversampling & Filtering – In-band Quantization Noise



$$P_{eq} = \frac{\Delta^2}{12} \frac{2}{f_s} f_b$$

$$= \frac{\Delta^2}{12} \frac{1}{OSR}$$

$$\frac{f_s}{2f_b} = \frac{f_s}{f_{\text{Nyquist}}} = OSR$$

"oversampling ratio (OSR)"

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Comparison – Nyquist vs. Oversampled ADC

$$SQNR_p = 10 \log_{10} \left(\frac{\frac{1}{2} \left(\frac{FSR}{2} \right)^2}{\frac{\Delta^2}{12} \frac{1}{OSR}} \right) = \left(\frac{\frac{1}{2} \left(\frac{2^B \Delta}{2} \right)^2}{\frac{\Delta^2}{12} \frac{1}{OSR}} \right)$$

$$SQNR_p = 6.02 \text{ dB} \cdot B + 1.76 \text{ dB} + 10 \log_{10} OSR$$

→ each bit B of the quantizer increases the resolution by 6 dB

→ each doubling of the OSR increases the resolution by 3 dB (half a bit)

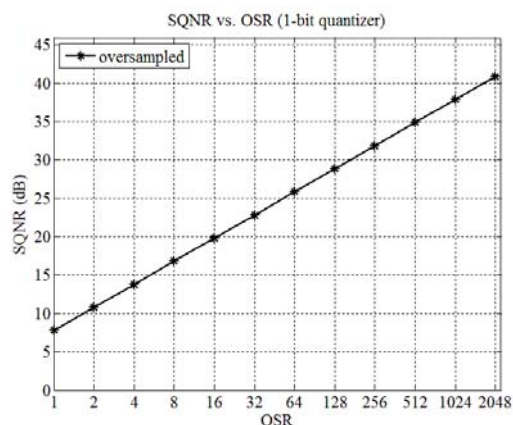
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SQNR versus OSR – 1-bit Quantizer



$SQNR \approx 42 \text{ dB}$ (7-bits) implies $OSR = 2048$

$f_b = 20 \text{ kHz}$ (Audio) → $f_s \approx 82 \text{ MHz}$

$f_b = 382 \text{ kHz}$ (GSM / EDGE) → $f_s \approx 1.6 \text{ GHz}$

$f_b = 1 \text{ MHz}$ (DSL) → $f_s \approx 4.1 \text{ GHz}$

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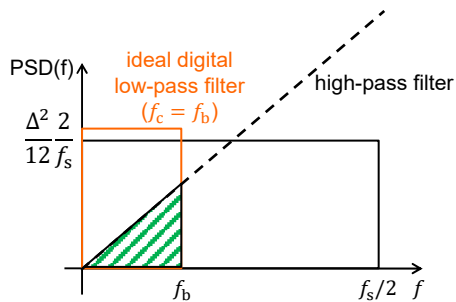
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Noise Shaping



high-pass filtering the quantization error before performing the digital low-pass filtering will result in a further reduction of the power in the signal band

→ “somehow” build an ADC that performs “noise-shaping”, i.e., reducing the power of the quantization error in the signal band while increasing it near $f_s/2$

Key: embed an ADC in a feedback loop!

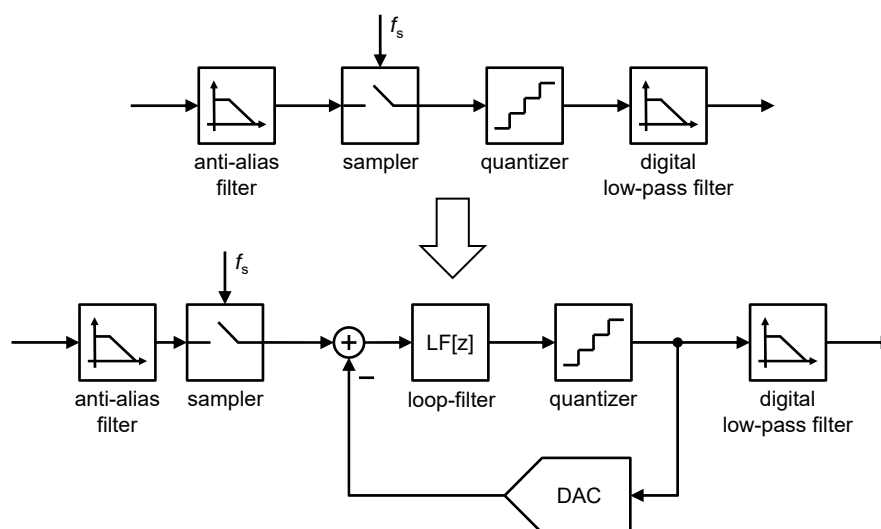
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From Oversampling to Noise-Shaping



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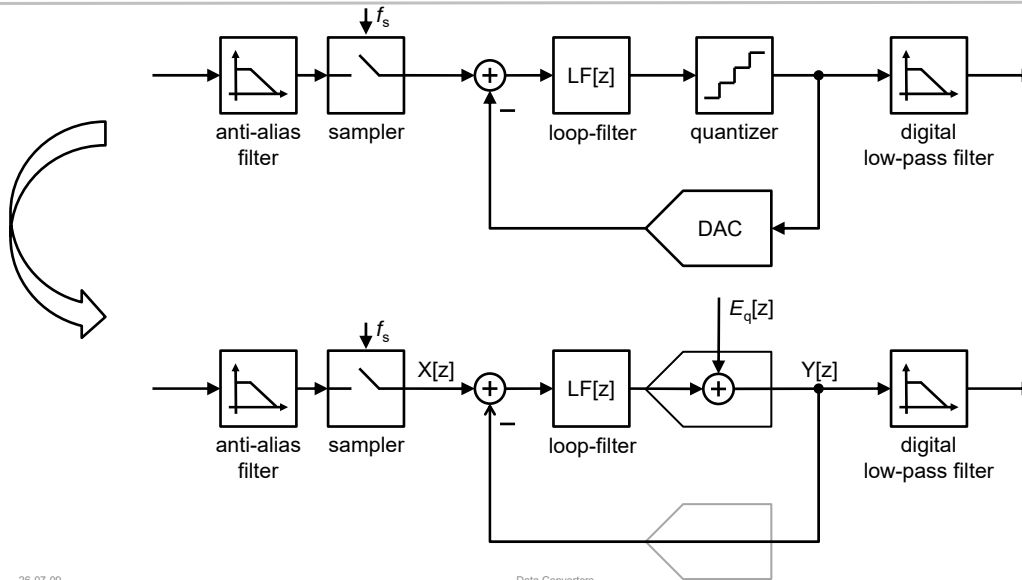
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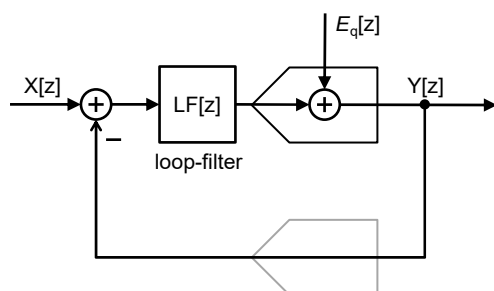
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Delta-Sigma ADC – Linear Model



Delta-Sigma Modulator – Linear Analysis



$$\begin{aligned}
 Y[z] &= E_q[z] + (X[z] - Y[z])LF[z] \\
 &= \frac{LF[z]}{1 + LF[z]} X[z] + \frac{1}{1 + LF[z]} E_q[z] \\
 &= \text{STF}[z] X[z] + \text{NTF}[z] E_q[z]
 \end{aligned}$$

signal noise
transfer function transfer function



Delta-Sigma ADC - Objectives

$$Y[z] = \underbrace{\frac{LF[z]}{1 + LF[z]}}_{\text{signal transfer function}} X[z] + \underbrace{\frac{1}{1 + LF[z]}}_{\text{noise transfer function}} E_q[z]$$

objectives:

$STF[z] \rightarrow$ unity in the signal band f_b

$NTF[z] \rightarrow$ small (ideally "zero") in the signal band f_b

for low-pass modulators, i.e., $0 \text{ Hz} \leq f \leq f_b$, this can be achieved

by making $|LF[z]| \gg 1$ at low frequencies

$\rightarrow STF[z] \approx 1$ & $NTF[z] \ll 1$

question: How to implement a loop-filter with this characteristic?

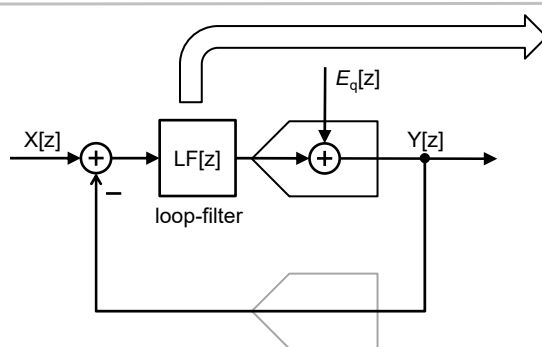
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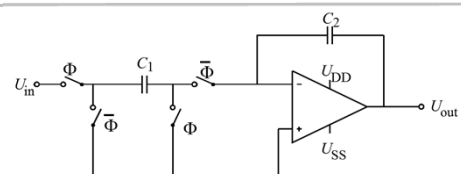
Implementing the Loop-filter – DT Integrator



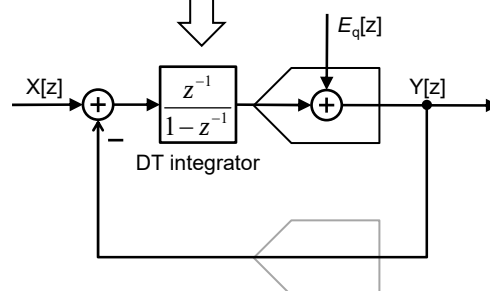
$$LF[z] = \frac{z^{-1}}{1 - z^{-1}} = \frac{1}{z - 1}$$

$$z = e^{-i2\pi f/f_s}$$

"infinite gain" at DC: $f = 0 \text{ Hz} \rightarrow z = 1$



switch-capacitor (SC) integrator
see "Microelectronics"



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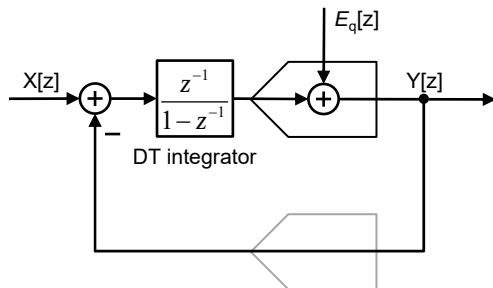
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First-Order Delta-Sigma Modulator



$$Y[z] = \frac{\frac{z^{-1}}{1 - z^{-1}}}{1 + \frac{z^{-1}}{1 - z^{-1}}} X[z] + \frac{1}{1 + \frac{z^{-1}}{1 - z^{-1}}} E_q[z]$$

$$= z^{-1} X[z] + (1 - z^{-1}) E_q[z]$$

delay
first-order high-pass filter

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NTF – Frequency Domain Analysis

$$NTF[z] = (1 - z^{-1})$$

note: $z = e^{-i2\pi f/f_s}$

$$NTF(f) = \left(1 - e^{-\frac{i2\pi f}{f_s}}\right)$$

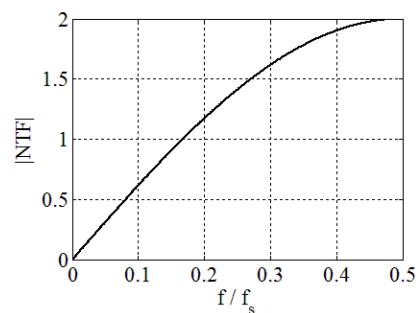
note: $\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$

$$= 2e^{-i\omega/(2f_s)} \frac{e^{i\omega/(2f_s)} - e^{-i\omega/(2f_s)}}{2}$$

$$= 2e^{-i\omega/(2f_s)} i \sin\left(\frac{\omega}{2f_s}\right)$$

$$= 2 \sin\left(\frac{\omega}{2f_s}\right) e^{-i\left(\frac{\omega}{2f_s} - \frac{\pi}{2}\right)}$$

$$|NTF(f)| = 2 \sin\left(\pi \frac{f}{f_s}\right)$$



quantization noise is attenuated at low frequencies and amplified at high frequencies
→ “first-order noise-shaping”

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1st-order Delta-Sigma ADC – In-band Noise

$$SQNR = 10 \log_{10} \frac{P_{\text{signal}}}{P_{\text{quantization noise}}}$$

$$\begin{aligned}
 P_{\text{quantization noise}} &= \int_0^{f_b} \frac{2\Delta^2}{12f_s} |NTF(f)|^2 df && \text{note: } |NTF(f)| = 2 \sin\left(\pi \frac{f}{f_s}\right) \\
 &= \int_0^{f_b} \frac{2\Delta^2}{12f_s} \left| 2 \sin\left(\pi \frac{f}{f_s}\right) \right|^2 df && \text{note: } \sin(x) \approx x \text{ for } x \ll \pi \\
 &\approx \int_0^{f_b} \frac{2\Delta^2}{12f_s} \left| 2\pi \frac{f}{f_s} \right|^2 df \\
 &= \frac{\Delta^2 \pi^2}{12 \cdot 3} \left(\frac{2f_b}{f_s} \right)^3 \\
 &= \frac{\Delta^2 \pi^2}{12 \cdot 3} \frac{1}{OSR^3}
 \end{aligned}$$

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1st-order Delta-Sigma ADC – SQNR

$$SQNR_p = 10 \log_{10} \left(\frac{\frac{1}{2} \left(\frac{FSR}{2} \right)^2}{\frac{\Delta^2 \pi^2}{12 \cdot 3} \frac{1}{OSR^3}} \right) = \left(\frac{\frac{1}{2} \left(\frac{2^B \Delta}{2} \right)^2}{\frac{\Delta^2 \pi^2}{12 \cdot 3} \frac{1}{OSR^3}} \right)$$

$$SQNR_p = 6.02 \text{ dB} \cdot B + 1.76 \text{ dB} - 5.2 \text{ dB} + 30 \log_{10} OSR$$

→ each bit B of the quantizer increases the resolution by 6 dB

→ each doubling of the OSR increases the resolution by 9 dB (1.5 bits)

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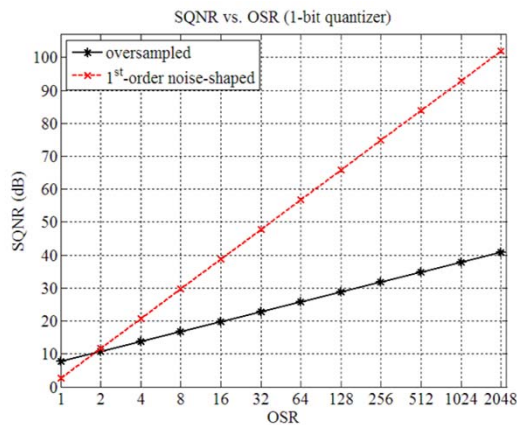
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Comparison



$SQNR \approx 42$ dB (7-bits) now implies $OSR \approx 21$

$f_b = 20$ kHz (Audio) $\rightarrow f_s \approx 840$ kHz

$f_b = 382$ kHz (GSM / EDGE) $\rightarrow f_s \approx 16$ MHz

$f_b = 1$ MHz (DSL) $\rightarrow f_s \approx 42$ MHz

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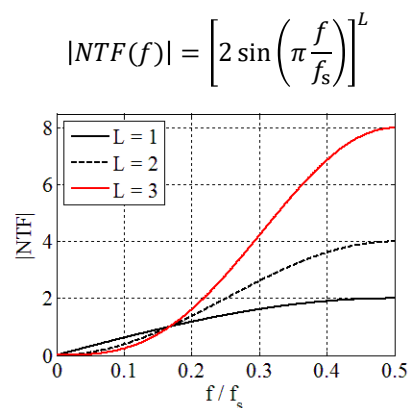
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What if...

... we were able to increase the order of the loop-filter to an arbitrary order L ?

$\rightarrow$ Let us assume that we were able to implement $NTF[z] = (1 - z^{-1})^L$

$$\begin{aligned}
 P_{\text{quantization noise}} &= \int_0^{f_b} \frac{2\Delta^2}{12f_s} |NTF(f)|^{2L} df \\
 &= \int_0^{f_b} \frac{2\Delta^2}{12f_s} \left| 2 \sin\left(\pi \frac{f}{f_s}\right) \right|^{2L} df \\
 &\approx \int_0^{f_b} \frac{2\Delta^2}{12f_s} \left| 2\pi \frac{f}{f_s} \right|^{2L} df \\
 &= \frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \left(\frac{2f_b}{f_s} \right)^{2L+1} \\
 &= \frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \frac{1}{OSR^{2L+1}}
 \end{aligned}$$



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L^{th} -order Delta-Sigma ADC – SQNR

$$SQNR_p = 10 \log_{10} \left(\frac{\frac{1}{2} \left(\frac{FSR}{2} \right)^2}{\frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \frac{1}{OSR^{2L+1}}} \right) = \left(\frac{\frac{1}{2} \left(\frac{2^B \Delta}{2} \right)^2}{\frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \frac{1}{OSR^{2L+1}}} \right)$$

$$SQNR_p = 10 \log_{10} \left(\frac{3}{2\pi^{2L}} 2^{2B} (2L+1) OSR^{2L+1} \right) = 10 \log_{10} \left(\frac{3\pi}{2} 2^{2B} (2L+1) \left(\frac{OSR}{\pi} \right)^{2L+1} \right)$$

$$SQNR_p = 6.02 \text{ dB} \cdot B + 10 \log_{10} \left(\frac{3}{2\pi^{2L}} (2L+1) \right) + (2L+1) \cdot 10 \log_{10} OSR$$

→ each bit B of the quantizer increases the resolution by 6 dB

→ each doubling of the OSR increases the resolution by $(2L+1) \cdot 3 \text{ dB}$

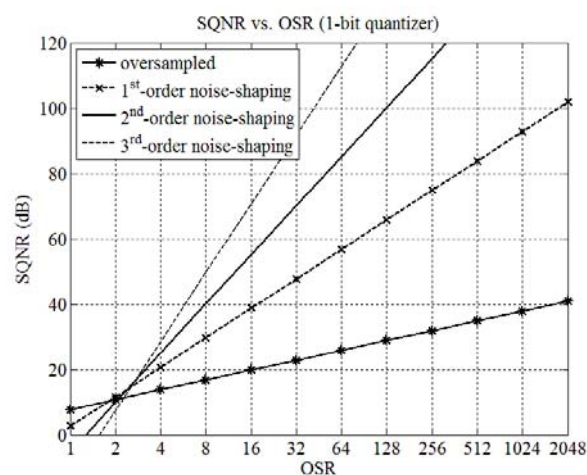
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Comparison



That looks very promising but...

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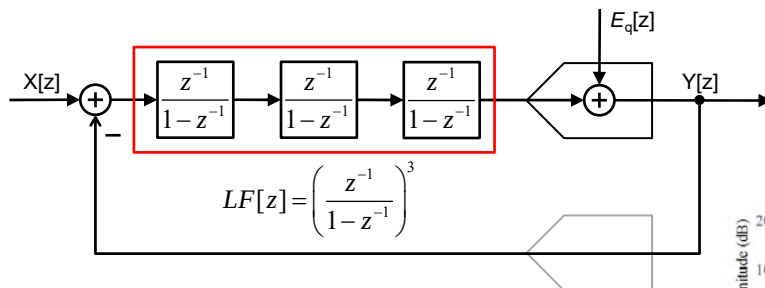
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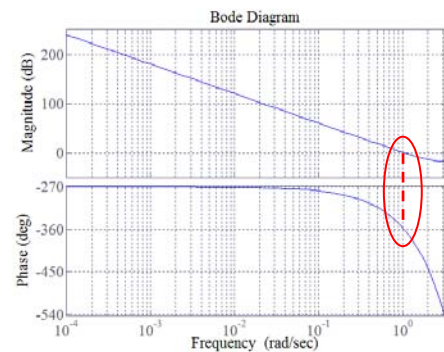


Delta-Sigma Modulator – Stability



$$LF[z] = \left(\frac{z^{-1}}{1-z^{-1}} \right)^3$$

It will and cannot not work this way!
→ stability!



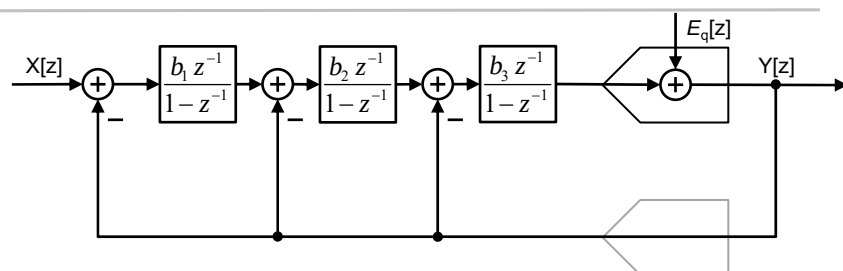
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Cascade of Integrators in Feedback (CIFB)



- solution: apply distributed feedback & scaling

$$LF[z] = b_1 b_2 b_3 \left(\frac{z^{-1}}{1-z^{-1}} \right)^3 + b_2 b_3 \left(\frac{z^{-1}}{1-z^{-1}} \right)^2 + b_3 \left(\frac{z^{-1}}{1-z^{-1}} \right)^1$$

$$= b_1 b_2 b_3 \left(\frac{1}{z-1} \right)^3 + b_2 b_3 \left(\frac{1}{z-1} \right)^2 + b_3 \left(\frac{1}{z-1} \right)^1$$

- 3 poles @ DC ($z = 1$)
- 2 zeros, adaptable by scaling coefficients

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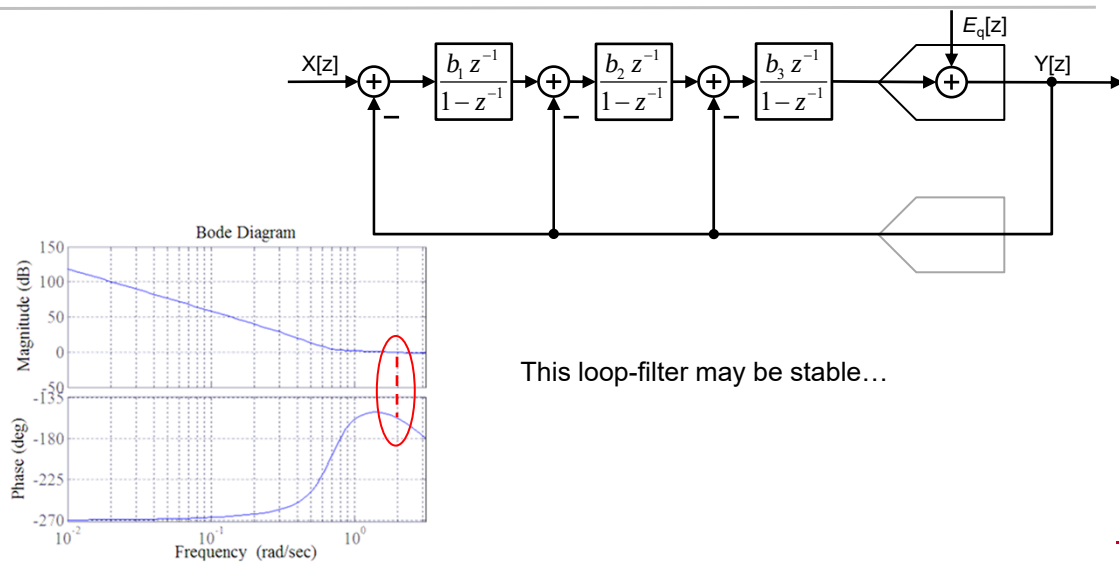
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CIFB - Stability



This loop-filter may be stable...



Impact of Scaling on Loop-filter

$$LF[z] = b_1 b_2 b_3 \left(\frac{z^{-1}}{1 - z^{-1}} \right)^3 + b_2 b_3 \left(\frac{z^{-1}}{1 - z^{-1}} \right)^2 + b_3 \left(\frac{z^{-1}}{1 - z^{-1}} \right)$$

Approximation within signal band, i.e., $f/f_s \ll 1$

$$LF[z] \approx b_1 b_2 b_3 \left(\frac{z^{-1}}{1 - z^{-1}} \right)^3$$

$$\begin{aligned} NTF[z] &= \frac{1}{1 + LF[z]} \\ &\approx \frac{1}{1 + b_1 b_2 b_3 \left(\frac{z^{-1}}{1 - z^{-1}} \right)^3} \\ &\approx \frac{1}{b_1 b_2 b_3} \left(\frac{1 - z^{-1}}{z^{-1}} \right)^3 \end{aligned}$$

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Impact of Scaling on Noise-shaping

$$\begin{aligned}
 P_{\text{quantization noise}} &= \int_0^{f_b} \frac{2\Delta^2}{12f_s} |NTF(f)|^{2L} df \\
 &= \int_0^{f_b} \frac{2\Delta^2}{12f_s} \frac{1}{(\prod_{i=1}^L b_i)^2} \left| 2 \sin\left(\pi \frac{f}{f_s}\right) \right|^{2L} df \\
 &\approx \int_0^{f_b} \frac{2\Delta^2}{12f_s} \frac{1}{(\prod_{i=1}^L b_i)^2} \left| 2\pi \frac{f}{f_s} \right|^{2L} df \\
 &= \frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \frac{1}{(\prod_{i=1}^L b_i)^2} \left(\frac{2f_b}{f_s} \right)^{2L+1} \\
 &= \frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \frac{1}{(\prod_{i=1}^L b_i)^2} \frac{1}{OSR^{2L+1}}
 \end{aligned}$$

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Impact of Scaling on SQNR

$$SQNR_p = 10 \log_{10} \left(\frac{\frac{1}{2} \left(\frac{FSR}{2} \right)^2}{\frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \frac{1}{\prod_{i=1}^L b_i^2} \frac{1}{OSR^{2L+1}}} \right) = \left(\frac{\frac{1}{2} \left(\frac{2^B \Delta}{2} \right)^2}{\frac{\Delta^2}{12} \frac{\pi^{2L}}{2L+1} \frac{1}{\prod_{i=1}^L b_i^2} \frac{1}{OSR^{2L+1}}} \right)$$

$$SQNR_p = 10 \log_{10} \left(\frac{3}{2\pi^{2L}} 2^{2B} (2L+1) OSR^{2L+1} \prod_{i=1}^L b_i^2 \right)$$

$$SQNR_p = 10 \log_{10} \left(\frac{3}{2\pi^{2L}} 2^{2B} (2L+1) OSR^{2L+1} \right) + 20 \log_{10} \prod_{i=1}^L b_i$$

Are these good or bad news?

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Impact of Scaling on SQNR - Comparison

	1-bit internal quantizer	4-bit internal quantizer
	product of scaling coefficients	
2 nd -order LF[z]	$1/2 \times 1/2 = 1/4$ (-12 dB)	$3/4 \times 9/4 = 27/16$ (+4.5 dB)
3 rd -order LF[z]	$1/5 \times 1/2 \times 1/2 = 1/20$ (-26 dB)	$1/2 \times 3/4 \times 9/4 = 27/32$ (-1.5 dB)
4 th -order LF[z]	$1/5 \times 1/5 \times 1/2 \times 1/2 = 1/100$ (-40 dB)	$1/4 \times 1/2 \times 3/4 \times 9/4 = 27/128$ (-13.5 dB)

$$\prod_{i=1}^L b_i < 1 \rightarrow 20 \log_{10} \prod_{i=1}^L b_i < 0 \text{ dB}$$

Conclusion:

Scaling must be performed for stability reasons although it (mostly) reduces $SQNR_p$!

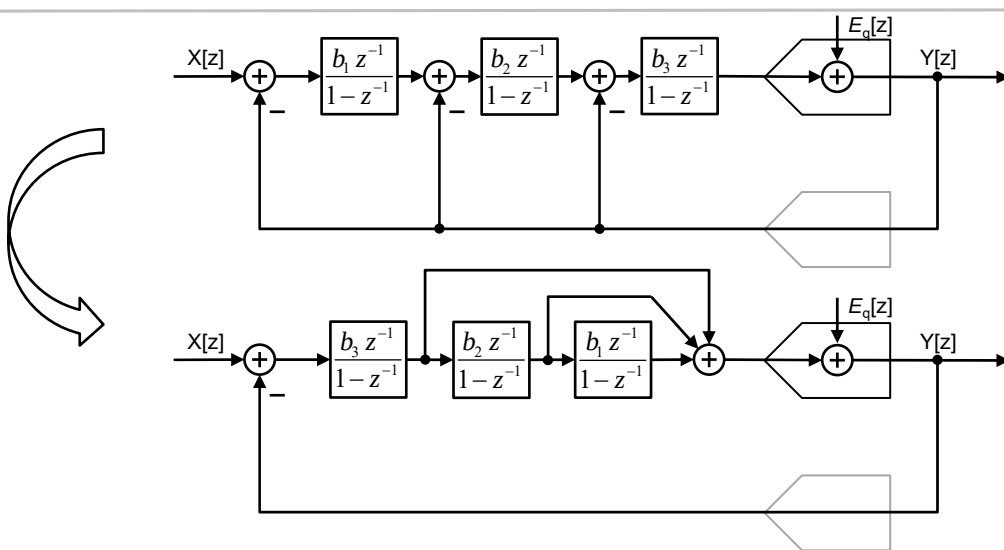
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Cascade of Integrators in Feed-forward (CIFF)



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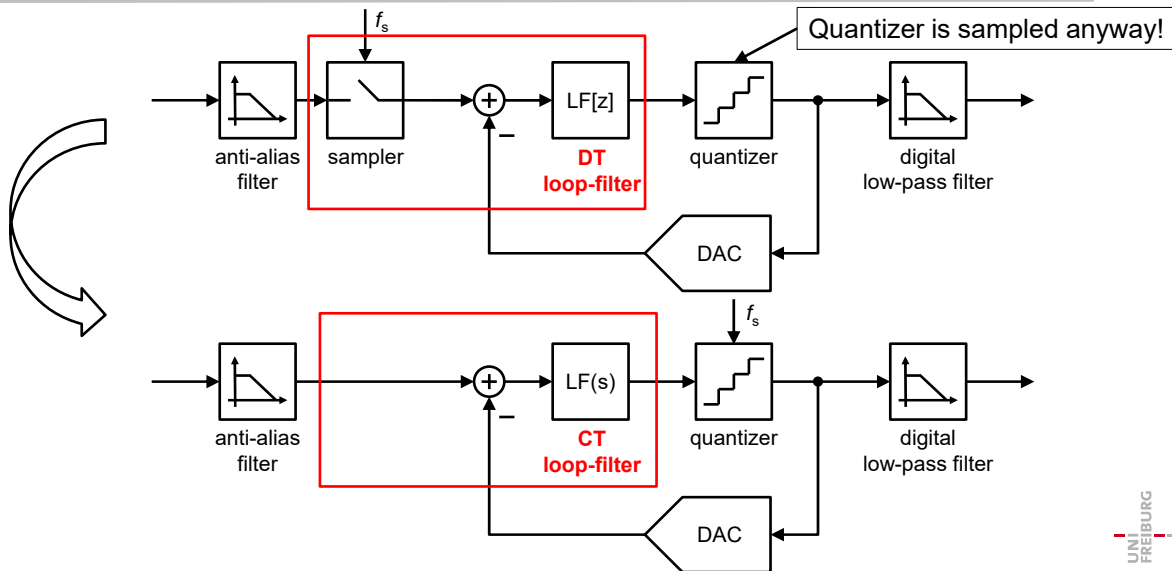
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From Discrete Time to Continuous Time



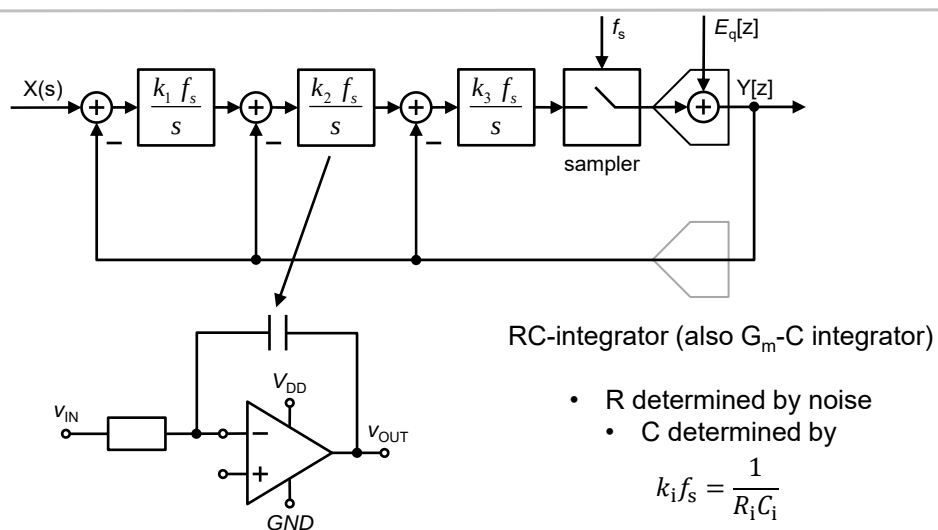
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Example – 3rd-order CT Delta-Sigma Modulator



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